

Azure Cloud Foundation

We mainly focused on three important Azure services:

Main Areas Covered

1. Compute

- **App Service** – Used to host web applications and APIs.
- **AKS (Azure Kubernetes Service)** – Used to run and manage containerized applications.

2. Storage

- **Blob Storage** – Used to store files and unstructured data.
- **Table Storage** – Used for storing structured NoSQL data.
- **Azure Files** – Used for shared file storage.

3. Identity

- **Microsoft Entra ID** – Used for identity and access management.

AWS and Azure Comparison

AWS	Azure
S3	Blob Storage
IAM	RBAC
EC2	App Service / Compute

Monitoring => Azure monitor

Data Protection

We also learned how Azure protects data using **MMK (Microsoft-managed keys)** for encryption.

Blob Storage – Mutable and Immutable Data

In Azure Blob Storage, data can be managed in two ways:

- **Mutable:** Blobs can be modified, overwritten, or deleted.
- **Immutable:** Data cannot be modified or deleted during the protection period. It can only be read or downloaded.

There are two main types of immutable storage:

- **Time-Based Retention:** Data is protected for a specific period of time.
- **Legal Hold:** Data remains protected until the legal hold is manually removed.

Creating an App Service

To create an App Service:

1. Select the **subscription** and **resource group**.
2. Enter the application name.
3. Choose how the application will be published.
4. Select the required **region**, such as Australia East.
5. Configure the database and container settings if required.
6. Configure the required ports and networking settings.
7. Review all the settings and click **Create**.

Azure App Service provides options such as:

- Web App
- Web App + Database
- WordPress on App Service
- Web App for Managed Instance

Creating Storage and Containers

To create Azure Storage:

1. Enter the basic details such as **name and region**.
2. Select the required **performance** and **redundancy** options.
3. Create a **container** and provide a name.
4. Configure the required encryption settings.
5. Review the configuration and click **Create**.

IAM / RBAC in Azure

Azure uses **Role-Based Access Control (RBAC)** to manage access to resources.

For example, to assign access to an App Service:

1. Open the App Service.
2. Go to **Access Control (IAM)**.
3. Select **Add Role Assignment**.
4. Choose the required role based on the user's responsibilities.
5. Select who should receive the access — a **user, group, or service principal**.
6. Select the required member/resource.
7. Review the settings and create the role assignment.

This helps ensure that users and applications only get the permissions they actually need.